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RESEARCH ARTICLE

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An evolutionary disruption of the buzz pollination syndrome in neotropical montane plants

Eduardo K. Nery[1,2] | **Mayara K. Caddah**[3] | **Fabian A. Michelangeli**[4] | **Anselmo Nogueira**[2]

[1]Programa de Pós-Graduação em Evolução e Diversidade, Centro de Ciências Naturais e Humanas, Universidade Federal do ABC, São Bernardo do Campo, SP, Brazil

[2]Laboratório de Interações Planta-Animal (LIPA), Centro de Ciências Naturais e Humanas, Universidade Federal do ABC, São Bernardo do Campo, SP, Brazil

[3]Departamento de Botânica, Universidade Federal de Santa Catarina, Florianópolis, SC, Brazil

[4]The New York Botanical Garden, Bronx, NY, USA

Correspondence

Eduardo K. Nery, Universidade Federal do ABC, Av. dos Estados 5001, Bairro Bangu, Santo André, SP, Brazil 09210-580.
Email: eduardo.k.nery@gmail.com

Abstract

Premise: Under pollinator limitations, specialized pollination syndromes may evolve toward contrasting responses: a generalized syndrome with increased pollinator attraction, pollinator reward, and pollen transfer capacity; or the selfing syndrome with increased self-pollen deposition, but reduced pollinator attraction and pollen transfer capacity. The buzz-pollination syndrome is specialized to explore female vibrating bees as pollinators. However, vibrating bees become less-active pollinators at montane areas of the Atlantic Forest (AF) domain. This study investigated whether the specialized buzz-pollination syndrome would evolve toward an alternative floral syndrome in montane areas of the AF domain, considering a generalized and the selfing syndromes as alternative responses.

Methods: We utilized a lineage within the buzz-pollinated *Miconia* as study system, contrasting floral traits between montane AF-endemic and non-endemic species. We measured and validated floral traits that were proxies for pollinator attraction, reward access, pollen transfer capacity, and self-pollen deposition. We inferred the evolution of floral trait via phylogenetic comparative methods.

Results: AF-endemic species have selectively evolved greater reward access and more frequently had generalist pollination. Nonetheless, AF-endemic species also have selectively evolved toward lower pollen transfer capacity and greater self pollination. These patterns indicated a complex evolutionary process that has jointly favored a generalized and the selfing syndromes.

Conclusions: The buzz pollination syndrome can undergo an evolutionary disruption in montane areas of the AF domain. This floral syndrome is likely more labile than often assumed, allowing buzz-pollinated plants to reproduce in environments where vibrating bees are less-reliable pollinators.

KEYWORDS

Atlantic Forest, generalization, macroevolution, Melastomataceae, pollinator limitation, self-pollination, specialization

Most angiosperms use their flowers to attract and reward animals that can transfer pollen from anthers to stigmas, i.e., pollinators (Armbruster, 2017). Pollinators have allowed long-distance pollen exchange and promoted outcrossing among flowering plants (Barrett, 2002; Goodwillie et al., 2005), thus enhancing offspring production and genetic variability. These fitness advantages have been presumably greater for plants whose floral traits could utilize specific pollinators groups (Johnson and Steiner, 2000; Brosi, 2016), making pollen transfer more accurate among conspecific plants (Moreira-Hernández and Muchhala, 2019). Consequently, different angiosperms lineages have evolved floral trait sets that can enforce pollinator specialization, i.e., specialized pollination syndromes (Fenster et al., 2004). Given their specialization, pollination syndromes have become reliable predictors of the most efficient pollinator group for angiosperm lineages (Rosas-Guerrero et al., 2014), highlighting a strong association between floral morphology and pollination system. However, pollinator specialization can make plants more prone to limited or unsuccessful outcrossing when their specific pollinators are less active or less numerous (i.e., pollinator limitations; Fenster and Martén-Rodríguez, 2007). In such cases, pollinator limitation would

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make specialized pollination syndromes unfit, allowing alternative floral syndromes to evolve in angiosperms.

A likely evolutionary response to pollinator limitations is a generalized pollination syndrome where floral traits allow cross pollination via multiple pollinator groups (Fenster et al., 2004). Albeit loosely defined, a generalized pollination syndrome often includes greater pollinator attraction, pollinator reward, and pollen transfer capacity (Fenster et al., 2004). Greater pollinator attraction and reward would ensure visitation by diverse groups of pollinators, increasing the chances of pollen removal and deposition (Armbruster, 2017). Consequently, greater pollen transfer capacity would allow these different pollinator groups to transfer pollen and also compensate for pollen loss from less-accurate pollinators (Harder and Johnson, 2023). These floral traits would maintain cross pollination when specific pollinator groups are limited, becoming an evolutionary alternative to specialized pollination syndromes (Johnson and Steiner, 2000). Illustrating this case, the primarily bee-pollinated Meriane tribe has evolved a mixed-vertebrate syndrome in tropical montane areas (Dellinger et al., 2019), where insects are less frequent pollinators (Dellinger et al., 2021). Hence, a generalized pollinator syndrome would be more robust to various limitations acting upon cross pollination, perhaps outperforming specialized pollination syndromes (Waser et al., 1996).

Another evolutionary response to pollinator limitation is the selfing syndrome where floral traits enhance self pollination (Sicard and Lenhard, 2011). The selfing syndrome often includes a greater capacity of self-pollen deposition, but with lower pollinator attraction and pollen transfer capacity (Sicard and Lenhard, 2011). A greater capacity for self-pollen deposition would allow plants to self-pollinate and produce offspring without receiving outcross pollen (Sicard and Lenhard, 2011), thus lessening pollinator dependency (Chomicki et al., 2020). However, as self pollination increases and ensures reproduction, pollinator attraction would become more costly for plants, selectively favoring lower pollinator attraction (Goodwillie et al., 2010). Similarly, pollen transfer capacity would become costly under lower frequencies of cross pollination, allowing smaller pollen transfer to evolve (Harder and Johnson, 2023). These traits would promote self pollination over cross pollination, likely disrupting specialized pollination syndromes (Cutter, 2019). For example, in the primarily bee-pollinated *Primula* genus, self-compatible species have evolved a greater proximity between anthers and stigma and smaller corollas (de Vos et al., 2014), indicating a greater capacity for self-pollen deposition at the expense of pollinator attraction. Hence, the selfing syndrome would indicate a mating system performing greater self pollination (Sicard and Lenhard, 2011), ensuring plant reproduction when pollinators become unreliable pollen vectors (Busch and Delph, 2012).

The buzz-pollination syndrome reflects a highly specialized pollination system in which cross pollination occurs via female bees from a specific group of species that can vibrate flowers (vibrating bees) (De Luca and Vallejo-Marín, 2013). Buzz-pollinated flowers greatly rely on their size to attract pollinators, and they often have pollen as the only reward type ("pollen" flowers) (Delgado et al., 2023). Once visited, these flowers will keep their pollen inside anthers that open via small pores and slits (poricidal dehiscence), which only release pollen under vibrations applied by bees (Vallejo-Marín, 2019). Hence, poricidal dehiscence restricts pollen presentation (Thomson et al., 2000), reducing access to reward and enforcing vibrating bees as pollinators. Nonetheless, the released pollen can be consumed by vibrating bees, representing a loss of male gametes (Vallejo-Marín et al., 2009). To lessen gamete loss, buzz-pollinated flowers often have two stamen morphs (heteranthery): a minor morph that allows pollen consumption by bees ("feeding" stamens); and a major morph that releases pollen over the bee body and hampers pollen consumption ("pollinating" stamens) (Vallejo-Marín et al., 2009). Consequently, the greater size of "pollinating" stamens indicates a relative investment in pollen transfer capacity over pollinator rewarding in heterantherous flowers (Vallejo-Marín et al., 2009). Despite this trade-off on pollen, the buzz-pollination syndrome has evolved in different angiosperm lineages (Buchmann, 1983), presumably reflecting a greater accuracy in pollen transference. Based on that, the buzz-pollination syndrome has been considered an "adaptive plateau" from which flower evolution would be unlikely to depart (Renner, 1989).

The Atlantic Forest (AF) is a neotropical domain where bee-pollinated plants can face increased pollinator limitations in montane areas. Bees have long been thought to be less-efficient pollinators in montane areas due to environmental conditions such as colder temperatures and rainy weather (Cruden, 1972). In supporting of this thought, bees and other hymenopterans are less common in the pollinator community in montane areas (Warren et al., 1988), and the proportion of plants species they pollinate decreases with elevation (Primack and Inouye, 1993). In tropical regions, bees become less common as pollinators at higher elevations (Dellinger et al., 2023). Following this pattern, bees also become less-active pollinators in montane areas of the AF domian. For instance, the buzz-pollinated *Tibouchina pulcra* receives three times less pollen in montane areas (elevation: 800–1100 m a.s.l.) than in lowland areas of the AF domain (Brito and Sazima, 2012). Hence, bee-pollinated plants are less fit in montane areas of the AF domain, where pollinator generalization and autogamy are relatively more frequent (e.g., Bergamo et al., 2021). Thus, the montane areas of AF domain can challenge the adaptive success of the buzz-pollination syndrome.

Here we used a lineage within the buzz-pollinated *Miconia* as a study system to investigate whether the specialized buzz-pollination syndrome evolved to an alternative floral syndrome in montane areas of the AF domain. We hypothesized that *Miconia* species have faced pollinator limitations in montane areas of the AF domain and consequently have undergone selective pressures favoring other floral syndromes. If so, we predicted two

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TABLE 1 Two floral syndromes that can evolve under pollinator limitations and their respective theoretical expectations over pollination processes. The dash symbol (—) indicates that a pollination process has no expectations under a given floral syndrome.

Pollination process	Generalized syndrome	Selfing syndrome
Pollinator attraction	increased	decreased
Pollinator reward	increased	—
Pollen transfer capacity	increased	decreased
Self-pollen deposition	—	increased

alternative evolutionary responses: (1) a generalized syndrome, where flowers have selectively evolved traits that allow cross pollination by multiple pollinator groups and (2) the selfing syndrome, where flowers have selectively evolved traits that enhance self pollination. Under a generalized syndrome, *Miconia* species in the AF domain should have evolved a greater pollinator attraction, pollinator reward, and pollen transfer capacity than *Miconia* species outside the AF domain (Table 1). Under the selfing syndrome, *Miconia* species in the AF domain should have evolved a greater self-pollen deposition capacity, but a smaller pollinator attraction and pollen transfer capacity, than *Miconia* outside the AF domain (Table 1).

MATERIALS AND METHODS

We conducted all analyses in the R environment v. 4.1.0 (R Core Team, 2021), but we only cited the main functions and their respective R packages. All data and R codes are available at GitHub (https://github.com/eknery/specialization_flower_AF).

Study area

Our study area was the Atlantic Forest (AF) domain (02°50′ S to 33°45′ S and 34°45′ W to 55°15′ W) in the Neotropics (IBGE, 2012). The AF domain is about 1,110,200 km^2 wide, ranging from sea level to approximately 2900 m a.s.l. Despite its lowland areas in the northernmost regions, the AF domain encompasses different mountain ranges and is considered a montane domain (Rahbek et al., 2019). The mountain ranges date from the Early Cretaceous to the Paleogene (145–23 million years ago [Mya]), influencing climatic zonation and soil formation within the AF domain (Hiruma et al., 2010; Alvares et al., 2013). These mountain ranges impose an elevation gradient that correlates with a reduction of pollen transference by bees (e.g., Brito and Sazima, 2012).

Study system

Our study system was the *Miconia* supersect. *Discolores* Caddah & Goldenberg (Melastomataceae), a lineage of woody flowering plants occupying the lowland Neotropics and montane areas of the AF domain (Caddah et al., 2022). Recent systematic studies have indicated 77 species in the *Miconia* supersect. *Discolores* (Caddah et al., 2022), of which 66 species (85%) have known phylogenetic placement (Nery et al., 2023). This group includes 23 species endemic to the AF domain (AF-endemic), 18 species spread between the AF and other domains (non-endemic), and 25 species outside the AF domain (non-endemic) (Nery et al., 2023). The elevational range of AF-endemic and non-endemic species has been previously characterized based on a taxonomic-curated dataset of occurrences gathered by Nery et al (2023). AF-endemic species occupy mostly montane forests (795 ± 144 m a.s.l., median ± interquartile range [IQR]), while non-endemic species occupy forests and open vegetation types at lower elevation areas (275 ± 678 m a.s.l.). Hence, AF-endemic species are overall closer to the elevational range where vibrating bees are less active in the AF domain (~800 m a.s.l.) (Brito and Sazima, 2012). Some non-endemic species also occupy montane areas outside the AF domain, mostly within open vegetation areas such as the *campos rupestres* (Caddah et al., 2022). Open vegetation areas are warmer and drier and should not hamper bee activity, presumably imposing little pollinator limitation to buzz-pollinated plants.

Like most melastomes, *Miconia* species have the buzz-pollination syndrome (Dellinger et al., 2022). Still, generalized pollination has been recorded in some species inside the AF domain (Brito et al., 2016). Their mating system can be either self-incompatible or self-compatible, and apomixis has been recorded in species outside the AF domain (Renner, 1989). Likely reflecting this variation in pollination and mating systems, *Miconia* species of our study system display different floral morphs inside and outside the AF domains (Goldenberg et al., 2008). These species have bisexual flowers with overall radial symmetry (Figure 1). The receptacle is elongated (hypanthium), raising the stamens and partially covering the pistil. The sepals are extremely reduced and fused, sometimes forming small calyx teeth. The petals are conspicuous, obovate-oblong in shape, and mostly white. The stamens are organized into an outer whorl ("pollinating" stamens) and an inner whorl ("feeding" stamens). Anthers open through apical pores and often have basal appendages, mostly reduced or absent in AF-endemic species. The pistils have straight styles ending into truncate or capitate stigmas, albeit a few non-endemic species can have styles curved away from anthers. Anther pores can be at stigma height or below it, thus characterizing reduced and reverse herkogamy, respectively (Opedal, 2018).

These *Miconia* species bloom across the year, peaking in the summer from December to February (M. K. Caddah, personal observations). Flowering is synchronous, and pollinator visits are more frequent in the morning (07:00–09:00 hours). Flowers often have anthers tilted upward at early anthesis, then downward during the day. Flowers usually last a few days, with petals decaying during the first day.

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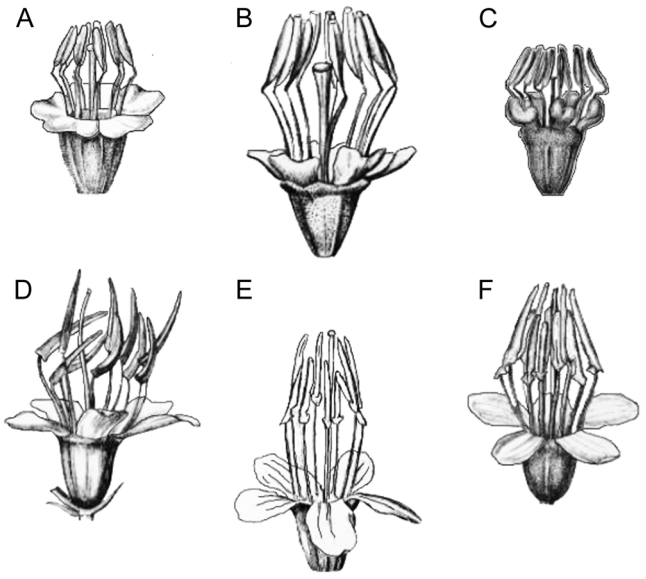


FIGURE 1 Flowers of species in the *Miconia* supersect *Discolores*. Flowers of AF-endemic species: (A) *Miconia lymanii* Wurdack, (B) *Miconia budlejoides* Triana, (C) *Miconia robusta* Cogn. Flowers of non-endemic species: (D) *Miconia stenostachya* DC., (E) *Miconia secundiflora* Cogn., (F) *Miconia heliotropoides* Triana. Illustrations adapted from Cogniaux (1888).

Measuring floral traits

To characterize the floral syndrome in our study system, we measured flowers on herbarium specimens representing species of the *Miconia* supersect. *Discolores*. Our sampling considered only expert-identified specimens in five herbarium collections (FLOR, MBML, MG, NY, and UPCB; Thiers, 2024). We sampled 151 specimens representing 51 species (3 ± 1 specimens per species), including 13 AF-endemic and 38 non-endemic species (Appendix S1). Non-sampled species were either rare or collected at the fruiting stage. We sampled specimens from different localities for each species to represent the intraspecific floral variation. We collected fully open flowers at apical (newer) inflorescence branches to lessen developmental differences. We rehydrated flowers and positioned their stigma and anthers upward to restore the flower arrangement during early anthesis. We photographed flowers with a ZEISS Stemi 305 stereo zoom microscope, capturing the lateral flower view and the ventral anther view. For each flower photograph, we used the program TPSdig (Rohlf, 2015) to measure pistil length (distance between the hypanthium base and the stigma), total stamen length (distance between the hypanthium base and the anther apex), anther length (distance between the anther base and apex), and vertical diameter of anther pore.

We then used these linear measurements to calculate four floral traits that are proxies for different aspects of pollination and mating systems. We used pistil length to quantify flower size, a proxy for pollinator attraction. Larger flowers are visually more appealing to pollinators and consequently more likely to receive pollinator visits (Dafni et al., 1997), such that large flowers are associated with increased outcrossing rates (Goodwillie et al., 2010). In buzz-pollinated flowers, larger flowers are visited by a greater diversity of vibrating bee species (Delgado et al., 2023). We considered pistil length as the most suitable measurement of flower size for *Miconia* species of our study system since their flowers are often laterally compressed in scorpioid or glomerulated inflorescences (Caddah et al., 2022). We acknowledge that our measurement only represents a visual dimension of pollinator attraction, which also includes floral shape, color, and chemical signaling.

We divided the vertical pore diameter by the anther length of the "feeding" stamen to quantify a pore-anther size (PAS) ratio, a proxy for reward access. Anther pore size regulates pollen presentation in buzz-pollinated flowers (Thomson et al., 2000), such that larger-pored flowers can release more pollen than smaller-pored ones (Kemp and Vallejo-Marín, 2021). In *Miconia* flowers, larger-pored anthers can release pollen by mechanical forces other than vibration, forecasting pollen access to other pollinators than vibrating bees (e.g., Goldenberg and Shepherd, 1998). Nonetheless, larger-pored *Miconia* flowers often have smaller anthers, setting a trade-off between pollen access and pollen quantity (Brito et al., 2016). Hence, our PAS ratio is similar to the ratio of pore area to anther length (PA/AL), which was first used to describe this trade-off in *Miconia* flowers (Brito et al., 2016). However, our PAS ratio better describes this allometric relationship between pore size and anther size by using the same physical dimension, length.

We calculated the difference in lengths between the "pollinating" stamen and the "feeding" stamen and divided it by the length of the "feeding" stamen to quantify stamen dimetrism, a proxy for the pollen transfer capacity. In heterantherous flowers, size differences between "pollinating" and "feeding" stamens indicate the relative investment in pollen transfer capacity over pollinator reward (Vallejo-Marín et al., 2009). Our measurement of stamen dimetrism describes this relative investment by indicating how many times the "pollinating" stamen was longer than the "feeding" stamen. Other measures of stamen dimetrism have been proposed and indicated a greater investment in the pollen transfer capacity in strictly outcrossing melastome lineages (Melo et al., 2021). Note that stamen dimetrism only describes the size dimension of heteranthery, which also includes shape and color differences.

We calculated the difference between the pistil length and the total length of stamens from the inner whorl to quantify herkogamy, a proxy for self-pollen deposition. Herkogamy describes the spatial separation between pollen presentation and pollen receipt within a flower and indicates how likely self-pollen deposition occurs (Opedal, 2018). Stamens below the stigma (approach herkogamy) are likely to hamper self-pollen deposition, whereas stamens above the stigma (reverse herkogamy) are likely to allow self-pollen deposition (Opedal, 2018). We

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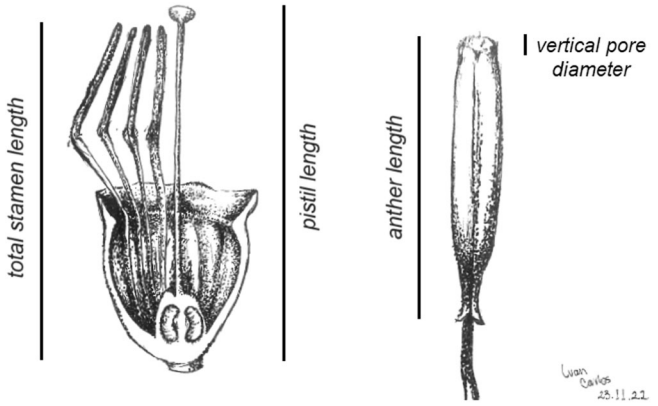


FIGURE 2 Schematic representation of *Miconia* flower showing traits measured. Illustrated by Luan Strombek.

considered our herkogamy measure suitable for *Miconia* species of our study system since their flowers hold "male" and "female" reproductive organs upward (Figure 2). This measure, however, would not be suitable for flowers in other plant lineages where herkogamy variation occurs in more than one spatial dimension.

Evaluating floral traits as proxies for pollination and mating system

To assess whether floral traits were reliable proxies for pollination and mating systems of *Miconia* species, we built a machine-learning model to explore associations between floral traits and pollinator generalization and self pollination in *Miconia* species. For this task, we assembled a data set including floral traits of *Miconia* species and information about their pollinator groups and frequency of self pollination.

We searched the literature to gather information about pollinator generalization and self pollination in *Miconia* species. We only considered species of *Miconia* s.s. (Goldenberg et al., 2008) that have flower morphology comparable to our study system. In our search of Google Scholar and Scopus platforms for relevant studies, we used the words 'Miconia' + 'pollin*', 'Miconia' + 'reprod*', 'Miconia' + 'general*', and 'Miconia' + 'self*' without filters for publication time. We also searched the seminal review by Renner (1989) for studies on the reproductive biology of *Miconia*. We considered studies with (1) field observations that identified the floral visitors and/or (2) field pollination experiments that quantified fruits after manual self-pollination (self-compatibility test) and after the exclusion of floral visitors (autogamy test). We gathered information on floral visitors and the percentage of autogamous fruiting for each *Miconia* species. We also gathered information on nectar production since nectar is also a pollinator reward that can affect pollination systems (Heil, 2011). We considered *Miconia* species as having nectar-producing flowers when information was available, otherwise assuming flowers were nectarless. We corrected taxonomic synonyms in our literature data set based on Goldenberg et al. (2013); hence, our literature search returned 15 studies containing information about 48 *Miconia* species, including 11 *Miconia* species from our study system (Appendix S2).

We assigned *Miconia* species in our literature data set to classes of pollination system and to continuous values of self pollination. Regarding the pollination system, any *Miconia* species visited only by bee species was classified as a specialist and as a generalist if visited by bee and non-bee species. We considered that visitation by non-vibrating bees would still support a specialized pollination system for *Miconia* species since these bees are common floral visitors, but without acting as effective pollinators. We observed that species classified as generalists were mostly visited by flies and rarely by vertebrates. For quantifying the frequency of self pollination, we used the percentage of fruit produce by autogamy. We acknowledge that autogamous fruiting can also arise from apomixis. To lessen this confounding effect, we excluded strictly apomitic species (25% of entries with mating information) from downstream steps. Hence, autogamous fruiting would reflect either self pollination and apomixis or only self pollination.

We utilized the MelastomaTRAIT database (Reginato et al., 2024) and a traditional taxonomic classification to gather floral trait values for *Miconia* species in our literature data set. The MelastomaTRAIT database provides mean values for traits measured for species of Melastomataceae. Trait values have been extracted from taxonomic descriptions by natural language processing or from datasets incorporated by contributing authors (Reginato et al., 2024). We utilized data from the MelastomaTRAIT database to calculate values of flower size, stamen dimetrism, and herkogamy for the *Miconia* species in our literature data set. However, the MelastomaTRAIT database does not include continuous-scaled values of anther pore size, precluding the calculation of the PAS ratio. Hence, we used the taxonomic classification by Cogniaux (1891) to assign discrete values of anther pore size to *Miconia* species. This taxonomic classification recognizes different sections within *Miconia* based on floral morphs defined in part by the relative size between the anther pore and the whole anther (Goldenberg et al., 2008). Following the classification of Goldenberg et al. (2013), we assigned *Miconia* species in our literature data set to the traditional sections (floral morphs) of Cogniaux (1891). We then used Cogniaux's (1891) sections to assign *Miconia* species to classes of anther pore size: (1) small-pored—species in sections *Adenodesma*, *Amblyarrhera*, *Hartigia*, *Jucunda*, *Laceraria*, *Miconia*, *Octomeris*, *Tamonea*; (2) large-pored—species in sections *Chaenanthera* (=*Hypoxanthus*), *Chaenopleura*, *Cremanium*, and *Glossocentrum*.

We built random forest (RF) models to assess whether floral traits could predict classes of pollination system and values of self pollination in *Miconia* species. We utilized the RF algorithm in the R package randomForest (Liaw and Wiener, 2002). The RF algorithms utilize known values from a set of variables (features) to create and weight

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different decision trees that can assign observations to unknown values of another variable (target) (Fawagreh et al., 2014). We built two models: (1) RF model of pollination system, with floral trait values as features and the classes of pollination system as the target; (2) RF model of mating system, with floral trait values as features and the frequency of self pollination as the target. We randomly split (seed = 41) the data set into a train set (60%) and a test set (40%). We utilized the train set to fit the RF model, and we utilized the test set to measure the predictive performance. We measured the predictive performance of the RF model of pollination system by the accuracy score (proportion of true positives), considering the Gini score to evaluate the importance of features. We evaluated the predictive performance of the RF model of mating system by the mean absolute error (MAE), considering the increase in node purity (INP) score to evaluate the importance of features. We compared the performance of our RF models to a dummy model whose predictions were a random draw of values from the target. We conducted RF fitting and testing by both including and excluding nectar production as a feature, so we could evaluate the effect of nectar on pollination and mating systems of *Miconia* species.

Finally, we utilized the RF models to assign classes of pollination system and values of self pollination to species of the *Miconia* supersect. *Discolores*. We retrained the RF models with our full literature data set, considering the following features: flower size, stamen dimetrism, anther pore size, and herkogamy. We utilized the continuous values of flower size, stamen dimetrism, and herkogamy measured on the herbarium specimens. We utilized the discrete value of anther pore size based on the floral morphs by Cougniaux (1891), as previously described. We utilized this approximation because the RF models were trained with discrete values of anther pore size, rather than with the PAS ratio. After assignments, we compared the relative frequency of pollination systems (generalist vs. specialist) in AF-endemic and non-endemic *Miconia* species, applying a χ^2 test of independence to assess differences between these groups. We also compared the frequency of self pollination in AF-endemic and non-endemic *Miconia* species, applying a permutation-based phylogenetic ANOVA to assess differences between these groups (Garland et al., 1993).

Phylogenetic hypotheses

To allow inferences about the evolution of floral traits, we used a sample of phylogenetic trees representing alternative evolutionary hypotheses for the *Miconia* supersect. *Discolores*. We retrieved a random sample ($N = 100$) of phylogenetic trees inferred by Nery et al. (2023), which provided the most robust phylogenetic reconstruction of the *Miconia* supersect. *Discolores* to date. The phylogenetic reconstruction applied a Bayesian framework over two nuclear and four plastidial loci representing 66 species of the *Miconia* supersect. *Discolores* and used two calibration points within *Miconia* for divergence time estimates. We also used the maximum clade credibility (MCC) of Nery et al. (2023) for illustrative purposes. We pruned phylogenetic trees to species whose flowers have been measured in our study.

Ancestral reconstruction of geographic distribution

To assess whether ancestral *Miconia* species were restricted to the AF domain and hence faced increased pollinator limitations, we conducted an ancestral reconstruction of geographic distributions in our sample of phylogenetic trees ($N = 100$). We reconstructed geographic distributions using the dispersal-extinction-cladogenesis (DEC) model (Ree and Smith, 2008) as implemented in the R package BioGeoBEARS (Matzke, 2013). We considered two geographic areas, AF and other domains, thus allowing for three possible geographic distributions: (1) AF, (2) AF and other domains, (3) other domains. We did not use time-structured dispersal rates since the AF domain is much older than our study system. We used the maximum marginal likelihood to assign ancestral species to a geographic distribution in each phylogenetic tree. We considered species in the AF geographic distribution as the AF-endemic and species in the other domains and in the AF and other domains distributions as the non-endemic.

Inference of floral trait evolution

To assess whether the evolution of floral traits responded to pollinator limitations in montane areas of the AF domain, we applied a model-fitting approach to floral traits of AF-endemic and non-endemic *Miconia* species of our study system. We used models based on the Brownian motion (BM) and the Ornstein–Uhlenbeck (OU) process, as implemented in the R package OUwie (Beaulieu et al., 2012). Under BM models, the phenotype changes as a function of a stochastic process with a certain rate (σ) (Felsenstein, 1988). Under OU models, the phenotype changes as a function of a stochastic process with a certain rate (σ), but also following a dragging process with a certain strength (alpha, α) toward a central value (θ) (Hansen, 1997). Biologically, BM models represent evolution under neutral processes, namely mutation and genetic drift, or under inconsistent selective pressures, while OU models represent evolution under stabilizing selective pressures or some sort of constraint (Beaulieu et al., 2012). So the model parameters would represent the evolutionary rate (σ), the optimum phenotypic value (θ), and selection strength (α) for each floral trait (Beaulieu et al., 2012).

We fitted seven evolutionary models to floral traits across our sample of phylogenetic tress ($N = 100$), considering AF-endemic and non-endemic species as under distinct evolutionary regimes of flower evolution. We fitted two BM-based and five OU-based models whose parameter

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values could be either fixed or variable between AF-endemic and non-endemic species (see Appendix S3 for detailed model description). We also fitted the White Nose (WN) model, under which phenotypic values derive from a single normal distribution, thus considering a non-evolutionary explanation for floral trait variation. We chose the best-fit model for a given floral trait in each phylogenetic tree by the corrected Akaike information criterion (AICc) (Sugiura, 1978). We considered a minimum difference of two AICc scores to choose a complex model over a simpler one. The model attaining the best-fit across most phylogenetic trees was considered the most suitable for a given floral trait. We retrieved parameter estimates of the most suitable model to assess the evolutionary scenario of floral traits.

We expected the OU-based models, i.e., selection-driven scenarios, to have the best fit to all floral traits to support our hypotheses. Nonetheless, we expected different estimates of model parameters to support each floral syndrome (Table 2). For a generalized syndrome, we expected greater θ values of flower size, PAS ratio, and stamen dimetrism for AF-endemic species than for non-endemic species, with both groups with similar θ values of positive (approach) herkogamy. For the selfing syndrome, we expected smaller θ values of flower size, stamen dimetrism, and herkogamy for AF-endemic species than for non-endemic species, with both groups with similar θ values for the PAS ratio.

RESULTS

Phylogenetic reconstruction indicated that the last common ancestor of the *Miconia* supersect. *Discolores* dated back to 8.1 Mya (95% HPD = 5.9–10.8 Mya) (Figure 3A). In this group, AF-endemic species evolved later than non-endemic species, emerging after two independent vicariance processes. More recent dispersion events among ancestral AF-endemic species led to novel lineages of non-endemic species.

In our study system, the flower size of AF-endemic species (5.6 ± 1.1, median ± IQR) was similar to that of non-endemic species (5.9 ± 2.7), but almost three times less variable (Figure 3B). The pore to anther size (PAS) ratio was 50% greater for AF-endemic species (0.12 ± 0.05) than for non-endemic species (0.08 ± 0.07) (Figure 3C). The stamen dimetrism was 37% smaller for AF-endemic species (0.15 ± 0.14) than for non-endemic species (0.24 ± 0.14) (Figure 3D). Herkogamy values were overall negative, i.e., indicating reverse herkogamy (Opedal, 2018), but the herkogamy value was 24% lower for AF-endemic species (–3.26 ± 0.51) than for non-endemic species (–2.62 ± 1.19) (Figure 3E).

TABLE 2 Two alternative predictions of floral syndrome evolution under pollinator limitation and their expected parameter values for floral traits measured on AF-endemic and non-endemic species in our study *Miconia* system.

	Generalized syndrome		Selfing syndrome	
Trait	AF-endemic	Non-endemic	AF-endemic	Non-endemic
Flower size	Greater θ	Smaller θ	Smaller θ	Greater θ
Pore-anther size (PAS) ratio	Greater θ	Smaller θ	Same for both, positive θ	
Stamen dimetrism	Greater θ	Smaller θ	Smaller θ	Greater θ
Herkogamy	Same for both, positive θ		Smaller θ	Greater θ

Floral traits predict pollination and mating systems

The random forest (RF) models indicated that floral traits could predict pollinator generalization and self pollination among the *Miconia* species in our literature data set. The RF model of pollination system (accuracy = 0.71) largely outperformed the dummy model (accuracy = 0.28), relying mostly on anther pore size (Gini = 1.35). Larger-pored anthers were more frequent in generalist species (83%) than in specialist species (15%). The RF model of mating system (mean absolute error [MAE] = 32.23) also outperformed the dummy model (MAE = 34.46), relying mostly on herkogamy (increase in node purity [INP] = 73) and stamen dimetrism (INP = 60), respectively.

The inclusion of nectar production improved both RF models. The RF model of pollination system achieved greater predictive performance (accuracy = 0.85) with the inclusion of nectar production (Gini = 0.46), but still relied mostly on anther pore size (Gini = 1.22). The RF model of mating system also achieved greater predictive performance (MAE = 30.59) with the inclusion of nectar production (INP = 38), relying mostly on stamen dimetrism (INP = 50).

Based on the assignments by the RF models, the frequency of pollinator generalization and self pollination differed between AF-endemic and non-endemic *Miconia* species of our study system. The generalist class of pollination system was about five times more frequent among AF-endemic species (61%) than among non-endemic species (13%) ($\chi^2 = 9.52$, df = 1, $P < 0.005$) (Figure 4A). The frequency of self pollination was also greater in AF-endemic species (64%) than in non-endemic species (58%) (pseudo-$F = 13.45$, $P < 0.05$) (Figure 4B).

Inference of floral trait evolution: OU-based models were overall the best fit

The OU-based models were the best-fit for three floral traits measured on *Miconia* species of our study system, indicating that flowers have evolved mostly under stabilizing selective pressures or some sort of constraint. The BMS model was the best-fit only for flower size, indicating a neutral evolutionary regime for this floral trait. The WN model did not outperform the best-fit evolutionary model.

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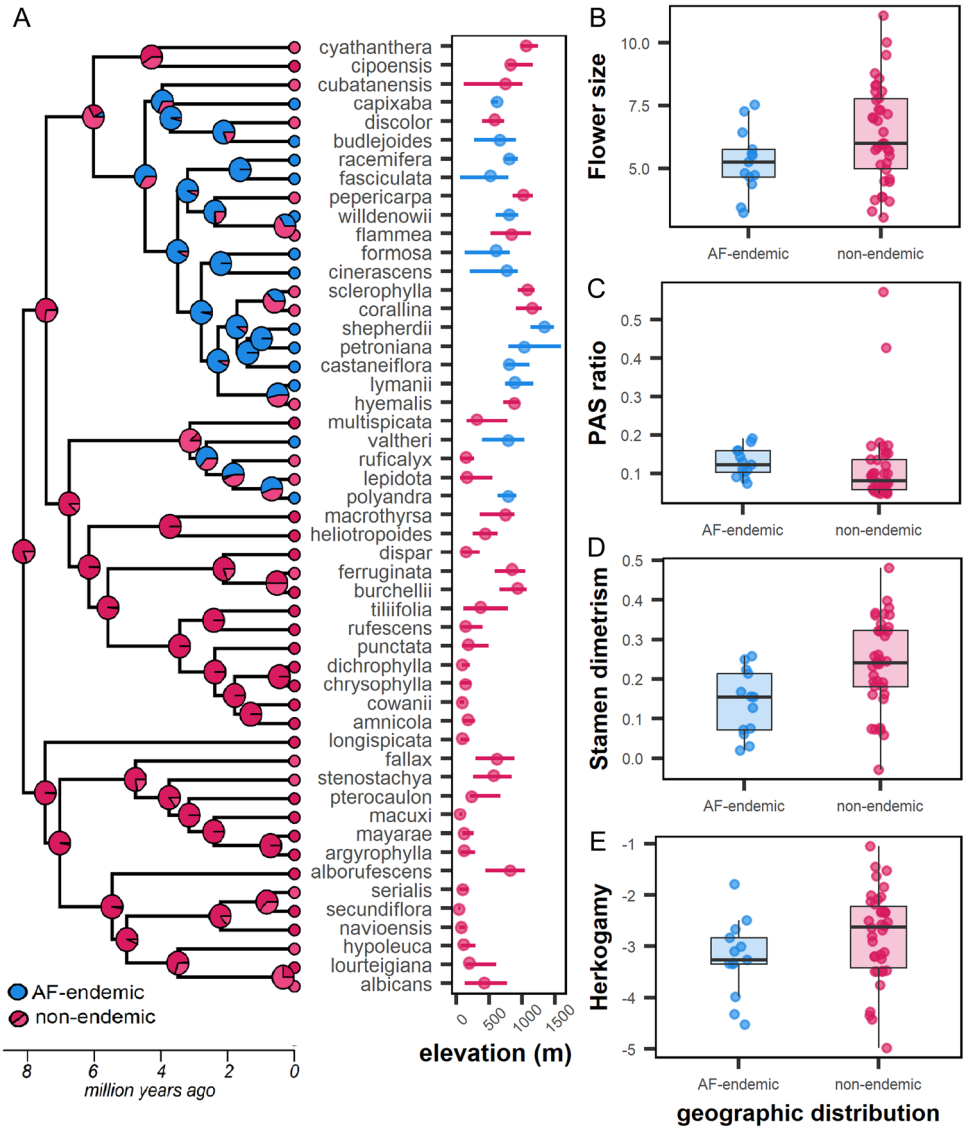


FIGURE 3 Phylogenetic relationships, elevation (m a.s.l.), and floral trait values for AF-endemic (blue) and non-endemic (red/pink) species in *Miconia* supersect. *Discolores*. (A) Phylogenetic relationships are based on the maximum clade credibility (MCC) phylogenetic tree. Ancestral geographic states inferred by the dispersal–extinction–cladogenesis (DEC) model. Dots and bars indicate the median and the 50% quantile range of elevation values occupied by species, respectively. Box plots display values for (B) flower size, (C) pore to anther size (PAS) ratio, (D) stamen dimetrism, and (E) herkogamy, all in millimeters.

The BMS model had the best-fit for flower size in 48% of phylogenetic trees, followed by the OUMV model in 39%. Under the BMS model, σ values differ between AF-endemic and non-endemic species. Sigma values for AF-endemic species ($\sigma = 2 \times 10^{-5} \pm 2 \times 10^{-6}$, mean ± SD) were much lower than those of non-endemic species ($\sigma = 0.51 \pm 0.05$) (Appendix S4), indicating that flower size of AF-endemic species evolved very slowly.

The OUMVA model had the best-fit for the pore to anther size (PAS) ratio in 68% of phylogenetic trees, followed by the OUMV model in 30%. Under the OUMVA model, all parameters differed between AF-endemic and non-endemic species. Theta values of AF-endemic species ($\theta = 0.132 \pm 0.016$) were 20% greater than those of non-endemic species ($\theta = 0.114 \pm 0.007$) (Figure 5A), indicating that AF-endemic species have evolved toward an optimum of larger-pored anthers. Sigma values of AF-endemic species ($\sigma = 0.02 \pm 0.01$) were eight times lower than those of non-endemic species ($\sigma = 0.17 \pm 0.05$) (Appendix S4), indicating that AF-endemic species have evolved slowly around their optimum of anther pore size. Alpha values of AF-endemic species ($\alpha = 1.52 \pm 0.81$) were 15% greater than those of non-endemic species ($\alpha = 1.32 \pm 1.00$) (Appendix S4), indicating that AF-endemic species have experienced a stronger selective pressure toward their optimum of anther pore size.

The OUM model had the best-fit for the stamen dimetrism in 97% of the phylogenetic trees, followed by the OUMV model in 3%. Under the OUM model, θ values

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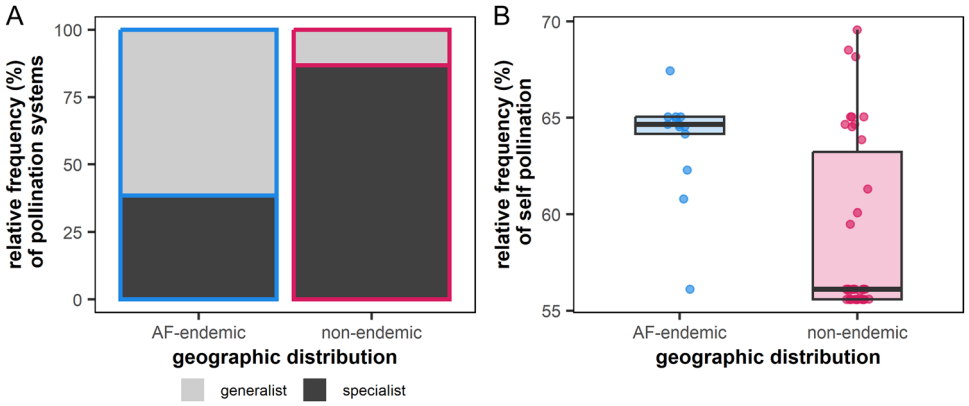


FIGURE 4 Random forest (RF) assignments of pollination system and self pollination between AF-endemic and non-endemic species in the *Miconia* supersect. *Discolores*. (A) Relative frequency of generalist and specialist pollination systems between AF-endemic and non-endemic species. (B) Relative frequency of self pollination between AF-endemic and non-endemic species. RF models utilized flower size, anther pore size, stamen dimetrism, and herkogamy as features.

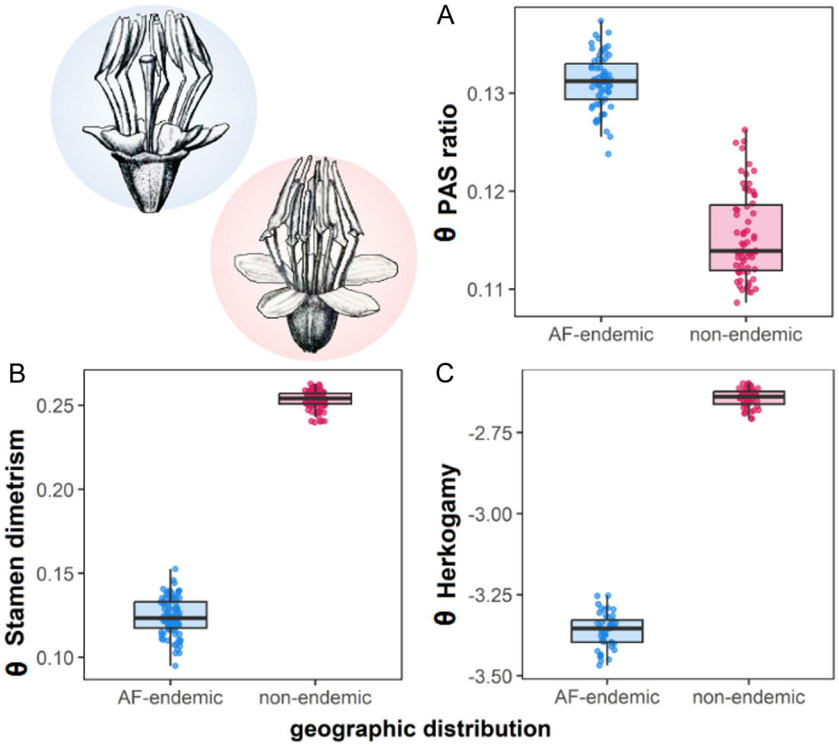


FIGURE 5 Theta (θ) estimates for three floral traits of AF-endemic and non-endemic species in the *Miconia* supersect. *Discolores* based on their best-fit evolutionary models. (A) Theta estimates for the pollen to anther size (PAS) ratio using the OUMVA model. (B) Theta estimates for stamen dimetrism using the OUM model. (C) Theta estimates for herkogamy using the OUM model.

differed between AF-endemic and non-endemic species, but σ and α values did not. Theta values of AF-endemic species ($\theta = 0.10 \pm 0.005$) were two times lower than those of non-endemic species ($\theta = 0.23 \pm 0.003$) (Figure 5B), indicating that AF-endemic species have evolved toward an optimum of lower stamen dimetrism (i.e., similar-sized stamens). Sigma estimates were relatively low ($\sigma = 0.11 \pm 0.09$) for AF-endemic and non-endemic species (Appendix S4), indicating that both AF-endemic and non-endemic species have evolved slowly near their respective optima of stamen dimetrism.

The OUM model had the best-fit for herkogamy in 44% of phylogenetic trees, followed by the OUMV model in 28%. Under the OUM model, θ values differed between AF-endemic and non-endemic species, but σ and α values did not. Theta values of AF-endemic species ($\theta = -3.36 \pm 0.05$) were 27% smaller than those of non-endemic species ($\theta = -2.64 \pm 0.03$) (Figure 5C), indicating

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that AF-endemic species have evolved toward an optimum of pronounced reverse herkogamy. Sigma values ($\sigma = 0.96 \pm 0.19$) were relatively high for AF-endemic and non-endemic species (Appendix S4), indicating that AF-endemic and non-endemic species have evolved rapidly near their optima of reverse herkogamy.

DISCUSSION

We investigated whether the specialized buzz-pollination syndrome would evolve toward alternative floral syndrome in montane areas of the AF domain, utilizing the buzz-pollinated *Miconia* supersect. *Discolores* as a study system. We hypothesized that *Miconia* species have faced greater limitations of vibrating bees in montane areas of the AF domain and consequently have undergone selective pressures favoring other floral syndromes. If so, we predicted two alternative evolutionary responses: (1) a generalized syndrome, where flowers have selectively evolved traits that allow cross pollination by multiple pollinator groups; and (2) the selfing syndrome, where flowers have selectively evolved traits that enhance self pollination. We found that AF-endemic species have selectively evolved toward larger-pored anthers and smaller stamen dimetrism than non-endemic species, supporting both the generalized syndrome and the selfing syndrome (Table 2). We also found an overall selective pressure toward reverse herkogamy, drawing a more complex scenario of floral syndrome evolution. These results indicate that the buzz pollination syndrome has undergone an evolutionary disruption in AF-endemic species of our study *Miconia* system, countering the perspective of an adaptive plateau (Renner, 1989).

Large-pored anthers support a generalized pollinator syndrome among AF-endemic species

Supporting a generalized syndrome (Table 2), AF-endemic *Miconia* species have selectively evolved toward an optimum of larger-pored anthers (greater PAS ratio), indicating increased reward access among these montane *Miconia* species. In buzz-pollinated flowers, poricidal dehiscence represents a mechanical restriction upon pollen presentation (Thomson et al., 2000), such that pollen release increases with anther pore size (e.g., Kemp and Vallejo-Marín, 2021). More specifically in *Miconia* flowers, larger-pored anthers are known to release pollen without vibrations by bees (Goldenberg and Shepherd, 1998). This mechanism would facilitate pollen release and access to other pollinators than vibrating bees, ultimately allowing a pollinator generalization of *Miconia* flowers (Brito et al., 2016). Supporting this case, our RF model indicated that larger-pored anthers could predict generalist pollination in *Miconia* species, also indicating a greater frequency of generalist pollination among AF-endemic *Miconia* species of our study system (Figure 4). Hence, the evolution of larger-pored anthers indicates a greater pollinator generalization among AF-endemic species of the *Miconia* supersect. *Discolores*.

Pollinator generalization would better allow plant species to maintain cross pollination when specific pollinator groups become less efficient (Waser et al., 1996). Bees have been known to become less-efficient pollinators in high-elevation areas (Cruden, 1972). This pattern holds across different montane areas (Primack and Inouye, 1993), intensifying in tropical regions (Dellinger et al., 2023). Following this pattern, vibrating bees can become less-efficient pollinators in montane areas of the AF domain (e.g., Brito and Sazima, 2012). Hence, the lower pollination efficiency of vibrating bees in montane areas of the AF domain has likely favored pollinator generalization among AF-endemic *Miconia* species of our study system. Supporting that possibility, generalist pollination systems are known to better cope with pollen limitation than specialist pollination systems in the AF domain (Wolowski et al., 2014). Likely reflecting this adaptive advantage, generalist pollination systems are more common than their specialist counterparts in plant communities of montane areas of the AF domain (e.g., Bergamo et al., 2021).

Shifts from a specialized to a generalized pollination syndrome are expected to involve changes in pollinator reward (Armbruster, 2017). Plants use floral rewards to energetically compensate animals for the pollination service (Bronstein et al., 2006). Still, reward types differ on their energetic budget and consequently can compensate different pollinator groups (Kooi et al., 2021). Hence, plants are more likely to explore novel pollinators when they can provide different reward types. For instance, in the Meriane tribe, floral nectaries are consistent markers of a mixed-vertebrate pollination syndrome that has evolved from the buzz-pollination syndrome (Dellinger et al., 2019), rewarding various vertebrate pollinators that are absent in classic buzz-pollinated flowers (Dellinger et al., 2021). Similarly, the evolution of floral nectaries can be a major trait underlying pollinator generalization among *Miconia* species, as indicated by the improved prediction of our RF model utilizing nectar production. For example, in our study system, *M. hyemalis* produces nectar and is visited by wasps and moths (Varassin et al., 2008), while *M. rufescens* is nectarless and is exclusively visited by vibrating bees (Renner, 1989). Nonetheless, nectar production and anther pore size may have confounding effects since most (76%) nectar-producing *Miconia* species had large-pored anthers in out literature data set. Hence, further experimental studies are needed to disentangle the effects of anther pore size and nectar production upon pollination systems of *Miconia* species.

Lower stamen dimetrism support the selfing syndrome among AF-endemic species

Supporting the selfing syndrome (Table 2), AF-endemic *Miconia* species have selectively evolved toward an optimum of lower stamen dimetrism, indicating reduced

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pollen transfer capacity among these montane *Miconia* species. In heterantherous flowers, the relative size between "pollinating" and "feeding" stamens indicates the relative investment in pollen transfer capacity (Vallejo-Marín et al., 2009). However, "pollinating" stamens can become costly for plants that can better ensure reproduction via self pollination that is presumably more accurate and demands less pollen (Harder and Johnson, 2023). Hence, under greater self pollination, smaller "pollinating" stamens would be more advantageous, consequently reducing the size difference (dimetrism) between stamen whorls. Regarding *Miconia* species, our RF model indicated that stamen dimetrism was a major predictor of self pollination, also indicating a greater frequency of self pollination among AF-endemic species of our study system (Figure 4). Similarly, stamen dimetrism has been previously inferred to selectively decrease in autogamous melastome lineages (Melo et al., 2021). Based on that, the evolution of lower stamen dimetrism indicates greater self pollination among AF-endemic species of the *Miconia* supersect. *Discolores*.

Nonetheless, lower stamen dimetrism reflects a consequence of greater self pollination (Melo et al., 2021), which is mediated by other floral traits. In our *Miconia* study species, different floral traits allow facilitated and autonomous self pollination. Regarding pollinator-facilitated selfing, reverse herkogamy allows pollen to drop from anthers to the stigma during vibrations applied by bees. Moreover, as flowers are compressed in glomerulated or scorpiod inflorescences (Caddah et al., 2022), large bees can vibrate several flowers at once, allowing self-pollen exchange within a same terminal inflorescence (F. A. Michelangeli, personal observations). Regarding autonomous selfing, larger-pored anthers allow self-pollen deposition by simple abiotic processes, mostly wind and rain. Goldenberg and Shepherd (1998) first observed that *Miconia* species with larger-pored anthers release pollen after a simple touch or brusque movement. Later, Brito et al. (2017) quantified self-pollen deposition in *Miconia* flowers with larger-pored anthers by counting pollen grains on stigmas of bagged flowers. Hence, the pronounced reverse herkogamy and the larger-pored anthers allow facilitated and autonomous self-pollen deposition in AF-endemic *Miconia* species of our study system, ultimately leading to greater self pollination (Figure 4).

Shifts toward greater self pollination can disturb specialized plant–pollinator interactions (Cutter, 2019). Since self pollination is more accurate and requires fewer pollen to attain reproduction, selfing-prone plants are more likely to evolve toward lower pollen quantities than their strictly outcrossing counterparts (Harder and Johnson, 2023). This reduction in pollen quantity could constrain the rewards to animals that cross-pollinate flowers (Bronstein et al., 2006). Reward constraints likely intensify in flowers offering pollen as the sole reward ("pollen" flowers), such as in the buzz-pollination syndrome (De Luca and Vallejo-Marín, 2013). Hence, the evolution of greater self pollination may lower pollen reward to vibrating bees, reducing the selective pressures maintaining the specialized buzz-pollinated flowers. In *Miconia*, the evolution of greater self pollination has likely led to flowers with shorter anthers, presumably lowering the pollen quantity of each flower. For instance, 80% of fruit set in *Miconia elaeoides* is from autonomous self pollination, and its anthers are shorter (1.6–2.9 mm long) (Manrique-Valderrama et al., 2022). Conversely, the strictly outcrossing *Miconia paucidens* has larger anthers (4–5 mm long) (Goldenberg and Shepherd, 1998). Intriguingly, *Miconia* species with shorter anthers often have larger pores (Brito et al., 2016), suggesting that greater self pollination often couples with generalist pollination, as previously discussed. Hence, greater self pollination could be an additional factor reducing pollinator specialization in buzz-pollinated *Miconia* flowers, a new hypothesis requiring further testing.

Flower size and reverse herkogamy indicate an overall trend of reduced pollinator dependency

Supporting neither floral syndrome (Table 2), AF-endemic and non-endemic *Miconia* species have evolved around the same optimum of flower size but different optima of reverse herkogamy. Flower size, our proxy for pollinator attraction, has evolved without any differences in phenotypic optima (Appendix S4), likely reflecting a neutral evolutionary regime (Beaulieu et al., 2012). Herkogamy, our proxy for self-pollen deposition, has evolved toward negative values in AF-endemic and non-endemic species, likely reflecting a selective pressure that favors self-pollen deposition in both groups (Beaulieu et al., 2012). Indeed, our RF model indicated relatively high frequencies of self pollination in both AF-endemic and non-endemic species. These patterns suggest an overall reproductive trend toward reduced pollinator dependency in *Miconia* species.

As noted already, flower size is a major component of pollinator attraction (Dafni et al., 1997) and positively correlates with outcrossing rates across angiosperm lineages (Goodwillie et al., 2010). Moreover, flower size determines pollinator diversity for buzz-pollinated flowers (Delgado et al., 2023). Nonetheless, in our study system, AF-endemic and non-endemic *Miconia* species evolved flower size under a neutral evolutionary regime, indicating reduced selective pressure on pollinator attraction in both groups. This reduced selective pressure on pollinator attraction can indicate a mating system with low dependence on pollinators for reproduction (Chomicki et al., 2020). Indeed, our RF model indicated self pollination as common for both AF-endemic and non-endemic *Miconia* species of our study system. This low pollinator dependency may be an ancestral, genus-level trait since *Miconia* species overall have smaller flowers than other "pollen" flower species, suggesting that pollinator attraction is reduced across the whole genus (Delgado et al., 2023). Thus, the neutral evolution of flower size indicates a reduced pollinator dependency across our *Miconia* study species, as further discussed.

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Herkogamy directly determines how likely self-pollen deposition occurs (Opedal, 2018). In reverse herkogamous flowers, self pollen can drop from anthers to the stigma due to pollinator handling or abiotic processes (Opedal, 2018). In our study system, both AF-endemic and non-endemic *Miconia* species have evolved toward reverse herkogamy, suggesting self pollen is deposited in both groups. Supporting the link between herkogamy and self-pollen deposition, our RF model indicated that herkogamy was the major predictor of self pollination in *Miconia* species, also indicating self pollination in both AF-endemic and non-endemic *Miconia* species of our study system (Figure 4). This trend toward reduced herkogamy has been recorded across other *Miconia* lineages (Gavrutenko et al., 2020), suggesting an overall increase in self pollination in this genus. An increase in self pollination has been thought to reflect frequent failure of cross pollination (Goodwillie et al., 2005), representing a reproductive assurance strategy (Busch and Delph, 2012). This reproductive assurance strategy can be mostly advantageous for highly specialized pollination systems that are presumably more prone to pollinator shortages (Fenster and Martén-Rodríguez, 2007). Hence, reverse herkogamy has evolved in *Miconia* species likely due to selective pressures favoring greater self pollination as a reproductive assurance strategy.

Notably, different *Miconia* species are known to produce fruits via apomixis (Renner, 1989), some without the need of pollen deposition (Caetano and Oliveira, 2022). Apomixis could thus have enabled *Miconia* species to reproduce without pollen exchange via vibrating bees, further reducing pollinator dependency. Consequently, apomixis would have influenced the evolution of *Miconia* flowers, possibly lessening the selective pressures to maintain the buzz-pollination syndrome. Testing such a hypothesis requires additional data to contrast flower evolution between apomictic and non-apomictic *Miconia* species. Regardless of the relevance of apomixis, our study supports that *Miconia* species are overall less dependent on vibrating bees for reproduction, contrasting with the notion of high pollinator dependency of buzz-pollinated flowers.

Evolutionary disruption of the buzz-pollination syndrome

In our study system, AF-endemic *Miconia* species have evolved a floral phenotype that allows pollination by multiple pollinators and greater self pollination, perhaps exemplifying a general trend among *Miconia* lineages. However, this floral phenotype does not fully fit a generalized syndrome or the selfing syndrome (Table 2). Such an intermediate outcome indicates that our rationale has oversimplified how a specialized pollination syndrome can evolve in response to pollinator limitations. On one hand, a generalized syndrome has been considered the opposing evolutionary pattern of specialized syndromes (Fenster et al., 2004), presuming a linear morphological gradient influenced by pollinator specificity. On the other hand, the selfing syndrome has been advocated as a common evolutionary alternative to pollinator-mediated floral syndromes (Sicard and Lenhard, 2011), presuming another linear morphological gradient influenced by pollinator dependency. These linear morphological gradients can be combined to define a planar floral morphospace responding to pollinator specificity and dependency. This floral morphospace derives from the theoretical perspective of Chomicki et al. (2020), who proposed specificity and dependency as two separate features of mutualistic interactions. Hence, we advocate that flower evolution of our *Miconia* species is better described as "steps" over this floral morphospace, rather than "jumps" among discrete floral syndromes.

Nonetheless, these evolutionary steps over the floral morphospace have changed key traits of the buzz-pollination syndrome in AF-endemic *Miconia* species of our study system. The buzz-pollination syndrome likely derives from selective pressures that have optimized pollen transference over pollen consumption in "pollen" flowers (De Luca and Vallejo-Marín, 2013). The small-pored or slit-opened anthers allow pollen access to vibrating bees mostly (Thomson et al., 2000), consequently lessening pollen consumption by other floral visitors. Such restrictive pollen access is often coupled with a relatively higher investment in pollen transfer via heterantery (Vallejo-Marín et al., 2010). In our study system, the AF-endemic *Miconia* species have evolved toward less-restrictive pollen access (larger-pored anthers) and reduced investment in pollen transference (shorter "pollinating" stamens), changing two key traits of the buzz-pollination syndrome. These changes likely occurred due to selective pressures favoring generalist cross pollinaton and greater self pollination in montane areas of the AF domain, where vibrating bees are less-reliable pollinators (Brito and Sazima, 2012). Thus, we here propose that the buzz pollination has undergone an evolutionary disruption in the AF-endemic species of the *Miconia* supersect. *Discolores*.

In studies on pollination systems, floral syndromes have been assumed to be surrogates for pollination systems (Fenster et al., 2004). Such an assumption relies on a predictive relationship between floral traits and effective pollinators across angiosperm lineages (Rosas-Guerrero et al., 2014). Here we validated this predictive relationship for *Miconia* species. However, this approach requires caution because it does not provide an empirical estimate of pollinator diversity and self fertilization, which requires data from fieldwork (Dellinger, 2020). In this sense, our work lacks data on the number of pollinator species interacting with each plant species (degree of generalization) and how many seeds each plant species produces via self pollination (selfing rates). Detailed fieldwork on our study species will likely reveal a more diverse and complex array of reproductive systems, likely reflecting differences in pollinator assemblage at the landscape and local scales. Still, floral syndromes will remain valuable morphological

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markers of a long-standing evolutionary history between angiosperms and their pollinators.

CONCLUSIONS

The buzz-pollination syndrome reflects a highly specialized pollination system that has been assumed to be very stable at the macroevolutionary scale (i.e., an adaptive plateau). However, pollinator limitations can make plant species with specialized pollination syndromes less fit and favor alternative floral syndromes, such as a generalized syndrome or the selfing syndrome. Here we hypothesized that buzz-pollinated *Miconia* species have faced greater pollinator limitations in montane areas of the Atlantic Forest (AF) domain, consequently evolving toward an alternative floral syndrome. We found that AF-endemic *Miconia* species have evolved greater capacity for rewarding pollinators and also have more generalist pollination systems. Moreover, AF-endemic *Miconia* species have evolved a reduced capacity for pollen transfer and also have a higher frequency of self pollination. Hence, AF-endemic *Miconia* species have evolved toward both a generalized syndrome and the selfing syndrome. These changes, nonetheless, have disrupted key traits of the buzz-pollination syndrome.

AUTHOR CONTRIBUTIONS

E.K.N conceived the study, collected, and analyzed the data and wrote the manuscript. M.K.C. provided taxonomic expertise for data collection and discussion. F.M. provided taxonomic expertise for data collection and discussion. A.N. conceived the study and wrote the manuscript.

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DATA AVAILABILITY STATEMENT

All data and R codes are available at GitHub (https://github.com/eknery/specialization_flower_AF).

ORCID

Eduardo K. Nery http://orcid.org/0000-0003-4636-1285
Mayara K. Caddah http://orcid.org/0000-0003-2285-208X
Fabian A. Michelangeli http://orcid.org/0000-0001-7348-143X
Anselmo Nogueira http://orcid.org/0000-0002-8232-4636

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix S1. Sampled specimens of species in the *Miconia* supersect. *Discolores*.

Appendix S2. *Miconia* species in our literature data set and information on floral morph, nectar production, floral visitors, and autogamous fruiting.

Appendix S3. The seven evolutionary models for floral traits of AF-endemic and non-endemic species in our study *Miconia* system.

Appendix S4. Parameter estimates of the best-fit evolutionary models for the floral traits measured on AF-endemic and non-endemic *Miconia* species.

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